

# BAYAPUREDDY NEERAJ KUMAR REDDY

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## Education

### Vellore Institute of Technology

Vellore, Tamil Nadu

B. Tech in Computer Science and Engineering - CGPA: 8.48

### Sri Chaitanya Institutions

Vijayawada, Andhra Pradesh

Intermediate (MPC) - Percentage: 84.0

### Sri Chaitanya School

Kadapa, Andhra Pradesh

High School - GPA: 9.8

## Experience

### Waterlabs.AI

June 2025 – July 2025

Machine Learning Intern

Bangalore, Karnataka, India

- Implemented an **NLP-based clustering system** leveraging **Sentence-BERT, UMAP and KMeans** to structure clinical notes, achieving strong semantic separation (**Silhouette: 0.8433**).
- Identified key clinical themes such as **Referrals, Critical Care, Billing, Recovery Notes etc.**, using unsupervised learning, achieving strong clustering performance (**DB: 0.5687, CH: 2430.53**) and visualized results with **2D UMAP** to improve medical note triaging and reduce manual categorization through clearly separated semantic groups.

## Projects

### Hostel Management System with Real-time Tracking

- Developed a **React-based frontend** with a smooth UI for **student tracking, attendance, and admin dashboards**.
- Implemented **role-based access** and secure navigation using **React Router** and **Redux** for state management.
- Developed and integrated full-stack functionality by building **RESTful APIs** using **Node.js** and **Express.js**, managing data with **MongoDB**, implementing **JWT-based authentication** for secure server-side operations.

### AI Research Agent for Competitive Drug Analysis

- Built an AI agent that takes a disease name as input and autonomously searches the web, extracts relevant drug information, and generates a **structured competitive analysis report**.
- Used **LangGraph** to design a **multi-step agent workflow** with **tool-calling capabilities** including real-time web search via **Tavily API** and LLM-powered report generation using **Groq**.
- Deployed the application on **Streamlit Cloud** enabling live, interactive demonstration of the **agent's reasoning** across drug comparison, market trends and business insights.

### Detecting Digital Fatigue Using Session-Level Machine Learning

- Processed **1.6 million user logs** by aggregating session behaviors such as product views, cart actions and purchases.
- Engineered **session-level features** while intentionally avoiding duration-based signals to prevent data leakage and improve model reliability.
- Addressed class imbalance using **SMOTE** and trained multiple classifiers, with **XGBoost** achieving a **92% F1-score**.

## Skills

**Languages:** C++, JavaScript, SQL, Python

**Frameworks & Libraries:** NumPy, Pandas, Matplotlib, Scikit-Learn, React.js, Node.js, Express.js

**Database Systems:** MySQL, MongoDB

**Tools & Technologies:** Git, Power BI, LangGraph, Data Analysis

**Coursework:** DBMS, Operating Systems, Computer Networks, OOPS, Machine Learning

**Soft Skills:** Teamwork and Collaboration, Analytical and Logical Thinking, Communication Skills

## Certifications

- AWS Cloud Practitioner – Amazon Web Services
- Machine Learning Specialization – Coursera
- AI Agents in LangGraph – DeepLearning.AI
- C++ Development Workshop – Shaastra, IIT Madras
- Artificial Intelligence Fundamentals – IBM SkillsBuild

## Social Engagements

**Core Member:** Radio Club, VIT

**Event Coordinator:** Sahiti TLA, VIT